

Bone Demineralization Disorders

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10/31/2024

Demineralization Disorders for Today

1. Osteoporosis/osteopenia

2. Osteonecrosis

3. Legg-Calve-Perthes

4. Osteomalacia

5. Rickets

6. Paget Disease

7. Osteopetrosis

8. Osteogenesis Imperfecta



Objectives: Osteoporosis and Osteopenia

- 1) Define osteoporosis and osteopenia
- 2) Know how to interpret a bone mineral density scan in order to diagnose osteoporosis
- 3) Identify the major cause of hip fractures
- 4) Discuss multiple ways to prevent hip fractures
- 5) Identify the gold standard medication for treatment of osteoporosis as well as it's efficacy



Objectives: Osteonecrosis, Legg-Calve-Perthes, Osteomalacia and Rickets

- 6) Identify major risk factors for osteonecrosis/Legg-Calve-Perthes and the gold standard for diagnosis and treatment based on age
- 7) Recognize the physical as well as radiological findings in osteomalacia and rickets
- 8) Discuss the differences and similarities between osteomalacia and osteoporosis



Objectives: Paget Disease, Osteopetrosis and Osteogenesis Imperfecta

- 9) Explain the pathophysiology, diagnosis and treatment of Paget Disease
- 10) Identify the etiology of osteopetrosis as well as the sequelae of this condition
- 11) Recognize the clinical presentation of osteogenesis imperfecta as well as the pathogenesis of this condition

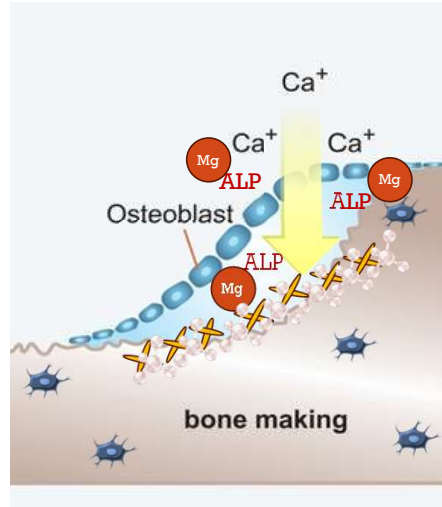


Bone Remodeling - Osteoblasts

Activation:

- Load bearing
- Hormones
- Diet

1. Lay down osteoid
2. Secrete alkaline phosphatase (ALP)
3. Mineralize bone
4. Become osteocytes



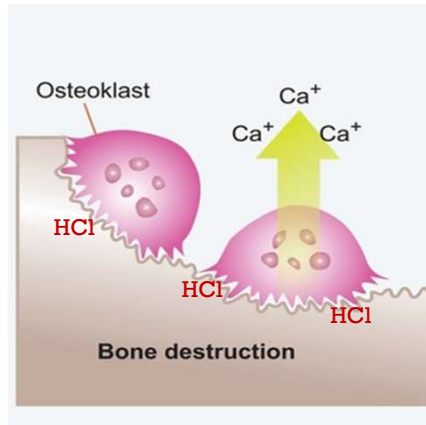
Alkaline phosphatase binds to the collagen in the osteoid to prepare it for mineralization

It provides phosphate for formation of hydroxyapatite

Magnesium is a cofactor for ALP

Role of magnesium in bone formation: Liu L, Luo P, Wen P and Xu P (2024) The role of magnesium in the pathogenesis of osteoporosis. *Front. Endocrinol.* 15:1406248. doi: 10.3389/fendo.2024.1406248

Bone Remodeling - Osteoclasts



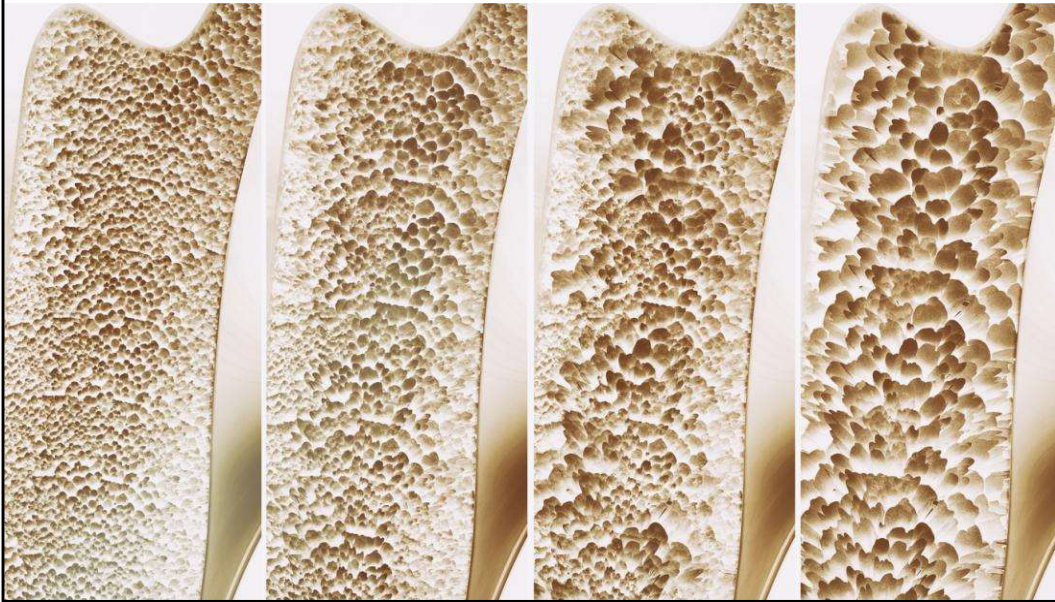
Activation:

- Inflammatory cytokines
- Damaged bone
- Low Calcium

Secrete HCl acid, collagenase and other enzymes to break down bone



OSTEOPOROSIS



Relatively normal on L and then progression

Why Osteoporosis Matters

The most common metabolic bone disease

40% of all women and 13% of all men in the United States will have an osteoporosis related fracture during their lifetimes

Impacts 1 out of 2 women and 1 out of 5 men

Can lead to significantly lower quality of life



Most common cause of non-traumatic back pain is osteoporosis – in study when treated the osteoporosis, pain improved.

Think about doing bone density screen when someone elderly comes in for back pain

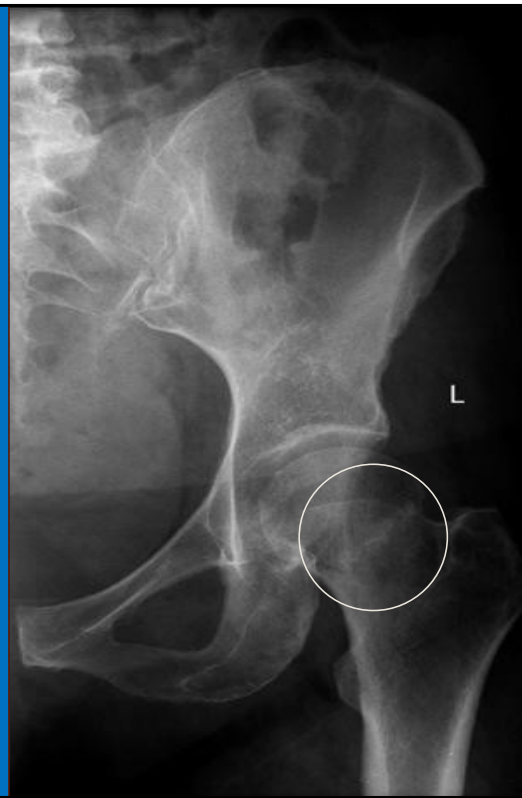
After a hip fracture

25% die in the first year

25% need long-term
nursing home care

25% will require someone to
assist them with ADLs for the
rest of their lives

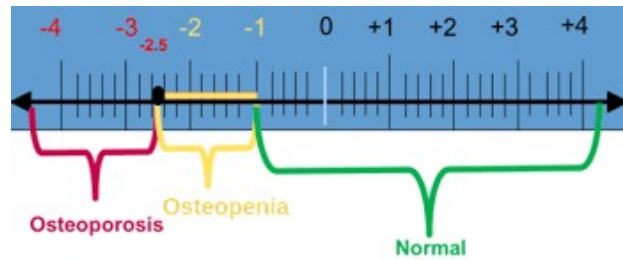
Less than 40% will get back to
prior functional status



Result of demineralization – fx at L femoral neck

After hip fx – 25% die in first year, 25% in nursing home, 25% will require someone to assist them with ADL for rest of life. Only 25% get back to prior fx status

DEFINITIONS



Osteoporosis is present when bone mineral density (BMD) is 2.5 standard deviations or more below the average value for young healthy women (a T-score of <-2.5 SD).

A second, higher threshold describes “low bone mass” or **osteopenia** as a T-score that lies between -1 and -2.5 SD.



Several definitions of osteoporosis. Look at normal density at age 30, when bone strongest. Then compare and see how much have lost compared to that. At 2.5, see higher fx risk

OSTEOPOROSIS

Primary Osteoporosis

Also called involutional osteoporosis

Secondary Osteoporosis

Medications (steroids, proton pump inhibitors, etc)

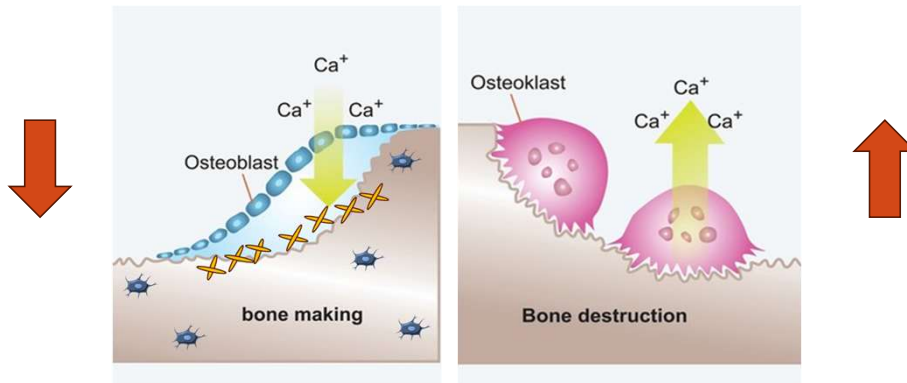
Endocrine disease (renal, parathyroid, etc.)

Lifestyle: alcohol and/or tobacco



OSTEOPOROSIS

Results from significant reduction in osteoblast to osteoclast activity



Osteoporosis is NOT a disease of calcium deficiency



First resting OB – get activation and then OC resorbs and then OB comes in and forms new bone

OSTEOPOROSIS

Same woman...
could that fracture
have been prevented?

Kyphosis decreases
lung capacity, blood
circulation and overall
life expectancy



Same woman, kyphosis in earlier image, also collar's fx – no one did a bone mineral density

Kyphosis decreases lung capacity, blood circulation and life expectancy

OSTEOPOROSIS RISK FACTORS

Modifiable

- Cigarette smoking
- Excessive alcohol use
- Poor diet
 - Need Ca, Vit D, Mg, Vit K...
 - Excess inorganic phosphorous
- Low body weight
- Low estrogen levels
- Medications

Not Modifiable

- History of fracture
- Age
- Family history
- Female
- Hyperparathyroidism
- Hyperthyroid
- Rheumatoid Arthritis
- Caucasian/Asian
 - Is this genetic?



Correcting hyperthyroid and paraHT does not correct risk of osteoporosis – we don't know why

Genetic risk:

Ethnic Variation in Bone Density in Premenopausal and Early Premenopausal Women: Effects of Anthropometric and lifestyle factors. July 2002

<https://doi.org/10.1210/jcem.87.7.8654>



Risk Factors - Medications

Increase osteoclast activity

- Glucocorticoids
- Estrogen depletion/receptor blockade (breast cancer)
- Depo Provera (reduces estrogen)

Decreased osteoblast activity

- Estrogen depletion/receptor blockade (breast cancer)
- Depo Provera (reduces estrogen)
- Androgen deprivation (prostate cancer)
- Diabetes medications
 - Pioglitazone (Actos)
 - Rosiglitazone (Avandia)
 - Thiazolidinedione's

Decreased calcium absorption

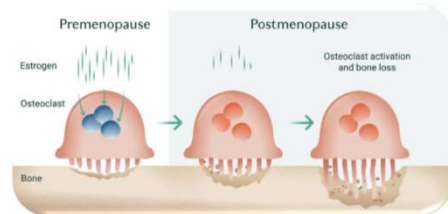
- Proton pump inhibitors

Increased calcium excretion

- Smoking (not a med!)
- Lasix

Prevent Vitamin D absorption

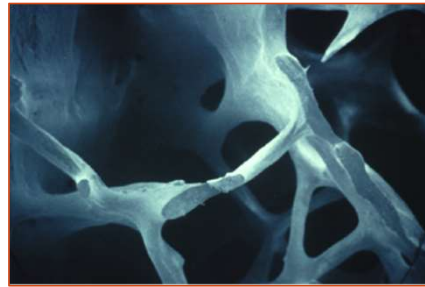
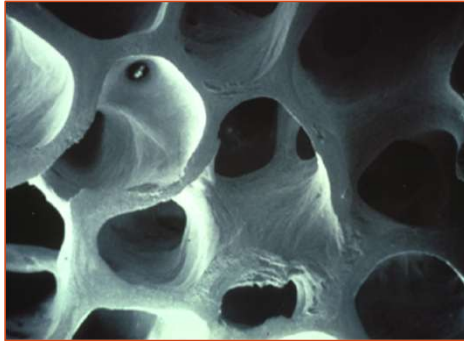
- Glucocorticoids



First column are factors that will increase risk of OP – second column are factors that will worsen existing OP

Note: not as well understood but PPIs also believed to cause aberrant osteoclast activity resulting in higher risk for OP

OSTEOPOROSIS



What is happening in osteoporosis – thinning of bone – small microfractures – get pain. Like a roof held up by a lot of columns – knocking out columns and finally get the large fx

SPINAL FRACTURE

The bone becomes more 'porous' or thinner



One side ultimately breaks down, causing it to become triangular



Kyphosis



As get thinner, one side breaks down, causing it to become triangular and then causing kyphosis

Dual Energy X-ray Absorptiometry (DEXA)



Dexa scan looks like.
Very low radiation and x-ray exposure

DIAGNOSTIC TESTING

Margin of error

DEXA	3 - 4%
Portable heel US	10%
Single Photon	8 - 10%
Dual Photon	8%
QCT	8%
X-ray	50%

LIMITATIONS

1. Can't be used to measure bone microstructure or quality, which are believed to influence fracture risk
2. Unable to differentiate trabecular versus cortical bone
3. DEXA is a 2D image so is not able to measure true volumetric BMD



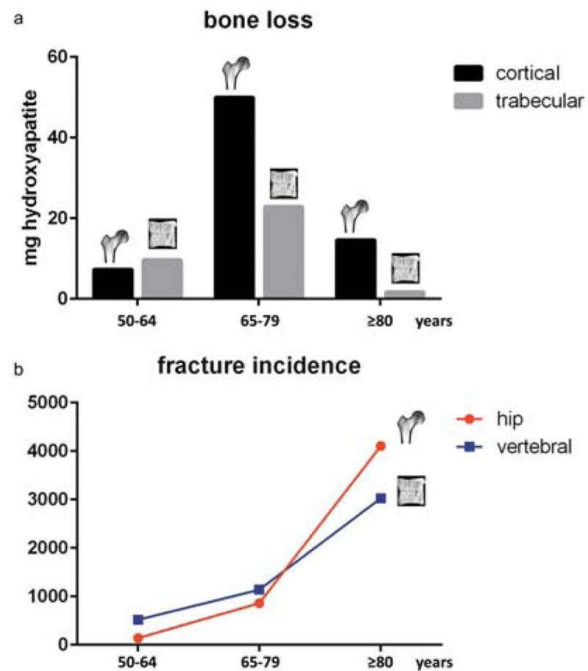
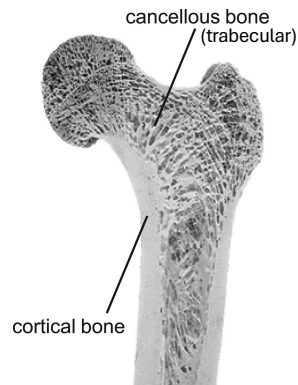
single energy X-ray beam. It is now widely considered inferior to dual-energy X-ray absorptiometry which uses a second energy beam to correct for absorption of X-ray energy by non-calcium containing tissues.

DEXA accurate to 3-4%. Gold standard.

Portable heel US machines can be off by about 10%.

Lorentzon, M., & Cummings, S. R. (2015). *Osteoporosis: the evolution of a diagnosis. Journal of Internal Medicine*, 277(6), 650-661.

Cortical Versus Trabecular Bone



Osterhoff G, Morgan EF, Shefelbine SJ, Karim L, McNamara LM, Augat P. Bone mechanical properties and changes with osteoporosis. *Injury*. 2016 Jun;47 Suppl 2(Suppl 2):S11-20. doi: 10.1016/S0020-1383(16)47003-8. PMID: 27338221; PMCID: PMC4955555.

We tend to lose more trabecular bone before the age of 65 and are more prone to vertebral fractures, where this type of bone predominates. After the age of 65, there is greater loss of cortical bone and an increase in hip fractures as the femur has more cortical bone

DEXA Screening (USPSTF)

Women age > 65 and Men age > 70

Women age 50-65 with risk factors or concern

Adults with a fracture after age 50

Anyone on steroid therapy

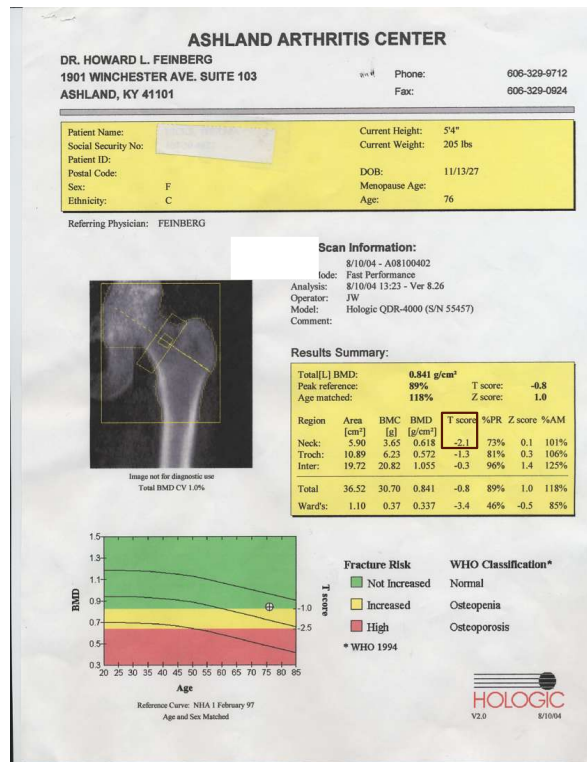


DEXA REPORT

Look for femoral
neck T-score

Osteopenia
Between -1 to -2.5

Osteoporosis
-2.5 or greater



Only need to look at femoral neck

On spine, need to have 2 consecutive T scores that are > -2.5 to have osteoporosis

Only thing you need is T-score.

DEXA REPORT

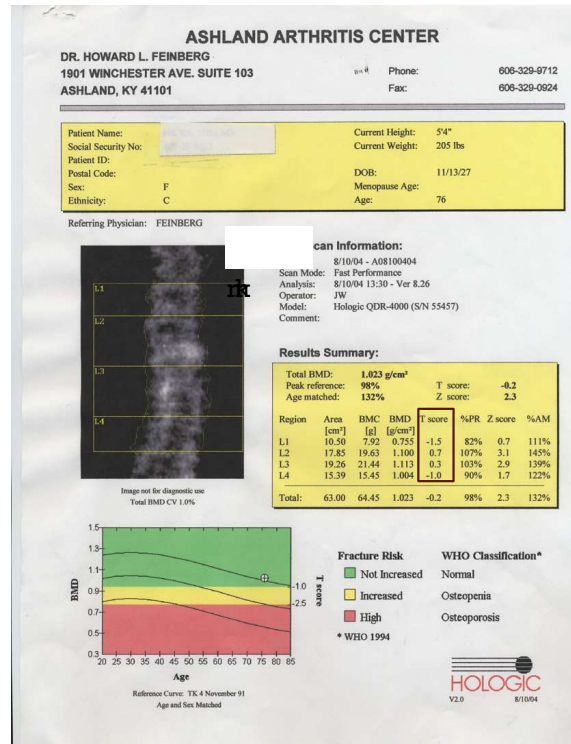
To diagnose osteoporosis based on a vertebral scan 2 consecutive T-scores must be abnormal

Osteopenia

Between -1 to -2.5

Osteoporosis

-2.5 or greater



Only need to look at femoral neck

On spine, need to have 2 consecutive T scores that are > -2.5 to have osteoporosis

Only thing you need is T-score.

OTHER TOOLS

QFRACTURE (fracture risk assessment tool)

- Externally validated

- Uses an extensive list of risk factors to predict 10-year risk

- Designed for the UK population

- Easy to understand results

Some clinicians initiate therapy early for high-risk patients

Unclear if pharmaceutical therapy has any benefit in preventing fractures. Can use risk status to mitigate modifiable risk factors and work on fall prevention



Osteoporosis dx

QFRACTURE has been externally validated. Designed for the UK population based on their risk outcomes. Easy to understand results

FRAX analysis – risk of fx over 10-yrs. Put on therapy if high

QFRACTURE.ORG

Welcome to the QFracture[®]-2016 risk calculator: <https://qfracture.org>

[Reset](#)
[Information](#)
[Publications](#)
[About](#)
[Copyright](#)
[Contact Us](#)
[Algorithm](#)
[Software](#)
[UKCA](#)

About you

Age (33-99):

Sex: ☐ Male ☒ Female

Ethnicity:

Clinical information

Smoking status:

Alcohol status:

Diabetes:

Do either of your parents have osteoporosis/hip fracture? ☐

Do you live in a nursing or care home? ☐

Have you had a wrist spine hip or shoulder fracture? ☐

History of falls? ☐

Dementia? ☐

Cancer? ☐

Asthma or COPD? ☐

Heart attack, angina, stroke or TIA? ☐

Chronic liver disease? ☐

Chronic kidney disease (stage 4 or 5)? ☐

Parkinson's disease? ☐

Rheumatoid arthritis or SLE? ☐

Malabsorption eg Crohn's disease, ulcerative colitis, coeliac disease, steatorrhea or blind loop syndrome? ☐

Endocrine problems eg thyrotoxicosis, hyperparathyroidism, Cushing's syndrome? ☐

Epilepsy or taking anticonvulsants? ☐

Taking antidepressants? ☐

Taking steroid tablets regularly? ☐

Taking oestrogen only HRT? ☐

Leave blank if unknown

Body mass index

Height (cm):

Weight (kg):

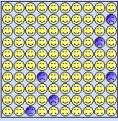
Your results

Your risk of having any osteoporotic (i.e. hip, wrist, shoulder or spine) fracture or hip fracture alone within the next 10 years is:

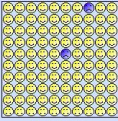
6.3%

1.9%

In other words, in a crowd of 100 people like you, 6 will develop osteoporotic fracture of hip, wrist, shoulder or spine within the next 10 years. Similarly, 2 will develop hip fracture within the next 10 years. This is represented by the smileys below:



fracture of hip, wrist, shoulder or spine



hip fracture

FRAX – limited to some risk factors. Has not shared validation data.

QFRACTURE has been externally validated. Designed for the UK population based on their risk outcomes. Easy to understand results

HIP FRACTURES

Most costly in terms of dollars and quality of life

Most hip fractures are NOT caused by osteoporosis, they are caused by FALLS

Our best data indicates low BMD is responsible for somewhere between 15-35% of ALL fractures



90% of all hip fractures result from falls: Daly RM, Dalla Via J, Duckham RL, Fraser SF, Helge EW. Exercise for the prevention of osteoporosis in postmenopausal women: an evidence-based guide to the optimal prescription. *Braz J Phys Ther.* 2019 Mar-Apr;23(2):170-180. doi: 10.1016/j.bjpt.2018.11.011. Epub 2018 Nov 22. PMID: 30503353; PMCID: PMC6429007.

Ha T Mai, Thach S Tran, Thao P Ho-Le, Jacqueline R Center, John A Eisman, Tuan V Nguyen, Two-Thirds of All Fractures Are Not Attributable to Osteoporosis and Advancing Age: Implications for Fracture Prevention, *The Journal of Clinical Endocrinology & Metabolism*, Volume 104, Issue 8, August 2019, Pages 3514–3520

Hadjimichael AC. Hip fractures in the elderly without osteoporosis. *J Frailty Sarcopenia Falls.* 2018 Mar 1;3(1):8-12. doi: 10.22540/JFSF-03-008. PMID: 32300689; PMCID: PMC7155350.

Järvinen TL, Michaëlsson K, Jokihaara J, Collins GS, Perry TL, Mintzes B, Musini V, Erviti J, Gorricho J, Wright JM, Sievänen H. Overdiagnosis of bone fragility in the quest to prevent hip fracture. *BMJ.* 2015 May 26;350:h2088.

Stone KL, Seeley DG, Lui LY, Cauley JA, Ensrud K, Browner WS, Nevitt MC, Cummings SR; Osteoporotic Fractures Research Group. BMD at multiple sites and risk of fracture of multiple types: long-term results from the Study of Osteoporotic Fractures. *J Bone Miner Res.* 2003 Nov;18(11):1947-54.

De Laet CE, van Hout BA, Burger H, Hofman A, Pols HA. *Bone density and risk of hip fracture in men and women: cross sectional analysis [published correction appears in BMJ 1997 Oct 11;315(7113):916]. BMJ.* 1997;315(7102):221-225

Centers for Disease Control and Prevention. *Hip Fractures Among Older Adults.* Accessed online October 24, 2023.

Emmerson BR, Varacallo M, Inman D. *Hip Fracture Overview.* StatPearls. Last updated August 2023. <https://www.ncbi.nlm.nih.gov/books/NBK557514/>

Fracture Risk is Multifactorial

Reduced bone mineral density

Weak muscles

Poor balance

What types of treatments could help minimize falls from each of these major risk factors?



Major risk factors include reduced bone density, weak muscles, poor balance. Exercise

Daly RM, Dalla Via J, Duckham RL, Fraser SF, Helge EW. Exercise for the prevention of osteoporosis in postmenopausal women: an evidence-based guide to the optimal prescription. *Braz J Phys Ther.* 2019 Mar-Apr;23(2):170-180. doi: 10.1016/j.bjpt.2018.11.011. Epub 2018 Nov 22. PMID: 30503353; PMCID: PMC6429007.

Preventing Falls and Fractures

Interventions can reduce number of falls as well as fall-related fractures by 40%

Exercise

Nutrition

- Dietary improvement, protein intake
- Calcium with Vitamin D? (more to come)

Management of urinary incontinence

Environmental assessment and modification

Assistive technology (walking aids, podiatry, etc.)

Correcting vision

Management of orthostatic hypotension

- Meds, rising slowly, etc



Dautzenberg L, Beglinger S, Tsokani S, et al. Interventions for preventing falls and fall-related fractures in community-dwelling older adults: A systematic review and network meta-analysis. J Am Geriatr Soc. 2021;69(10):2973-2984

Sherrington C, Fairhall NJ, Wallbank GK, et al. Exercise for preventing falls in older people living in the community. Cochrane Database Syst Rev. 2019;1(1):CD012424

Preventing Falls – Exercise is a Key Factor

Reduces falls as well as fractures significantly

- Meta-analyses suggest somewhere between 20-60%
- Women benefit more than men

Weight-bearing activities

stimulate osteoblasts and collagen deposition

All activity increases muscle strength/improve balance

Tai-chi has been shown to improve outcomes

Combination resistance, aerobic and balance training can reduce fall risk by over 70%

Exercise must be tailored to patient – safety and preference



Dautzenberg L, Beglinger S, Tsokani S, et al. Interventions for preventing falls and fall-related fractures in community-dwelling older adults: A systematic review and network meta-analysis. *J Am Geriatr Soc.* 2021;69(10):2973-2984

Sherrington C, Fairhall NJ, Wallbank GK, et al. Exercise for preventing falls in older people living in the community. *Cochrane Database Syst Rev.* 2019;1(1):CD012424

Karinkanta S, Kannus P, Uusi-Rasi K, Heinonen A, Sievänen H. Combined resistance and balance-jumping exercise reduces older women's injurious falls and fractures: 5-year follow-up study. *Age Ageing.* 2015;44(5):784-789

Wong RMY, Chong KC, Law SW, Ho WT, Li J, Chui CS, Chow SKH, Cheung WH. The effectiveness of exercises on fall and fracture prevention amongst community elderlies: A systematic review and meta-analysis. *J Orthop Translat.* 2020 Jun 1;24:58-65. doi: 10.1016/j.jot.2020.05.007.

Karinkanta S, Kannus P, Uusi-Rasi K, Heinonen A, Sievänen H. Combined resistance and balance-jumping exercise reduces older women's injurious falls and fractures: 5-year follow-up study. *Age Ageing.* 2015;44(5):784-789.

USPSTF Calcium and Vitamin D Guide

Clinical Summary: Vitamin D, Calcium, or Combined Supplementation for the Primary Prevention of Fractures in Community-Dwelling Adults

Population	Men and premenopausal women	>400 IU of vitamin D and >1000 mg of calcium in postmenopausal women	≤400 IU of vitamin D and ≤1000 mg of calcium in postmenopausal women
Recommendation	No recommendation. Grade: I (insufficient evidence)	No recommendation. Grade: I (insufficient evidence)	Do not recommend. Grade: D

Risk Assessment	Low bone mass, older age, and history of falls are major risk factors for incident osteoporotic fractures. Other risks factors for low bone mass and fractures include female sex, smoking, use of glucocorticoids, and use of other medications that impair bone metabolism (eg, aromatase inhibitors). Absolute fracture risk is very low in premenopausal women compared with postmenopausal women.
Preventive Medication	The recommendation against supplementation at lower doses was based on an overall assessment that supplementation at low doses provides no benefit. Evidence on the effect of supplementation on fractures at higher doses is conflicting, with some studies showing a reduction in certain fractures at higher doses and others showing no reduction or even an increase. More studies are needed to more clearly determine if supplementation with vitamin D, calcium, or both consistently prevents fractures. If future evidence shows a benefit, the magnitude of that benefit will need to be weighed against the magnitude of harms caused by supplementation (kidney stones).
Other Relevant USPSTF Recommendations	The USPSTF recommends against vitamin D supplementation to prevent falls in community-dwelling adults 65 years or older. The USPSTF recommends exercise interventions to prevent falls in community-dwelling older adults at increased risk for falls; multifactorial interventions may also be effective in some persons as well. The USPSTF recommends screening for osteoporosis in women 65 years or older and in younger women at increased risk. The USPSTF concludes that the current evidence is insufficient to assess the balance of benefits and harms of screening for vitamin D deficiency in asymptomatic adults.

USPSTF site:

<https://www.uspreventiveservicestaskforce.org/uspstf/recommendation/vitamin-d-calcium-or-combined-supplementation-for-the-primary-prevention-of-fractures-in-adults-preventive-medication>

Cochrane review: Méndez-Sánchez L, Clark P, Winzenberg TM, Tugwell P, Correa-Burrows P, Costello R. Calcium and vitamin D for increasing bone mineral density in premenopausal women. Cochrane Database of Systematic Reviews 2023, Issue 1. Art. No.: CD012664. DOI: 10.1002/14651858.CD012664.pub2. Accessed 24 October 2023

Bottom Line

Current USPSTF guidelines show mixed evidence on calcium and vitamin D supplementation in preventing fractures in community-dwelling individuals

Who should receive supplementation?

1. Patients living in nursing homes
2. Patients with osteoporosis
3. Dosing is not well established
 - 600-800 IU Vitamin D daily
 - 1000-1200mg calcium total intake daily

Data is mixed – does seem to ↑ BMD but not ↓ fractures



Ask for limitations in mobility/pain/prior fx to catch possible osteoporosis

Pharmaceutical Treatment

Bisphosphonates are first line

- Inhibits osteoclasts and osteoclast precursors to slow the rate of bone resorption
- For every 100 women taking bisphosphonates for 3 years, 5 vertebral and 1 hip fracture prevented
- Make sure calcium and vitamin D levels normal before starting
- Usually treat for 5 years and stop due to possible osteonecrosis of the jaw and atypical fractures
- Examples: alendronate, risedronate, ibandronate

Selective estrogen receptor modulator (SERM)

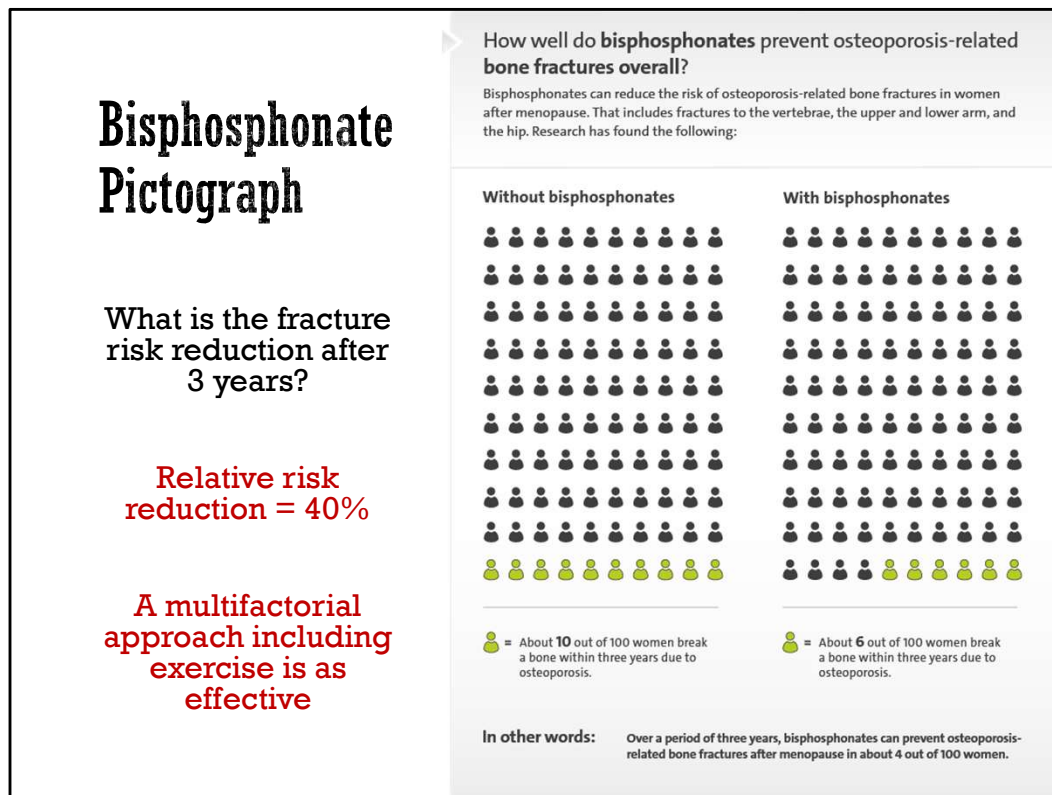
- Raloxifene
- Slows bone resorption, decreases risk of vertebral fractures
- Like HRT, increased risk for VTE

Estrogen therapy is NOT indicated for osteoporosis



<https://thennt.com/nnt/bisphosphonates-for-fracture-prevention-in-postmenopausal-women-with-prior-fractures-or-very-low-bone-density/>

UpToDate, Overview of the management of osteoporosis in postmenopausal women, accessed 10/24/23



Billington E, Aghajafari F, Skulsky E, Kline G

A. Bisphosphonates *BMJ* 2024; 386 :e076898 doi:10.1136/bmj-2023-076898.

(Diagram showing benefit with 10 years of bisphosphonate tx)

3-yr data from independent German non-profit:

<https://www.informedhealth.org/what-are-the-pros-and-cons-of-medication-to-prevent-bone-fractures.html>

Do We Treat Osteopenia?

T-score between
-1 and -2.5

Current guidelines recommend

- Healthy diet, including adequate calcium and vitamin D
- Weight bearing physical activity
- Smoking cessation
- Moderate to no alcohol consumption
- Avoidance of medication that can decrease BMD

Currently, pharmacological therapy is not recommended. Some providers will start medications for patients at high risk of future fracture



OSTEONECROSIS

Many names

Avascular Necrosis

Ischemic necrosis

Aseptic necrosis

Bone death

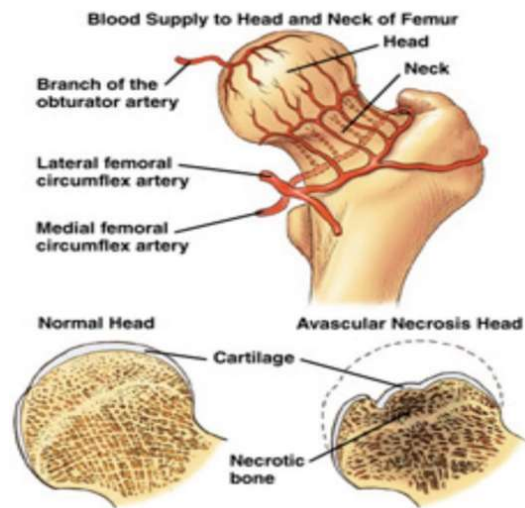


Image Credit: Oluwatoyin Olatundun Ilesanmi

OSTEONECROSIS EPIDEMIOLOGY

Estimated 20,000 cases a year in the U.S.

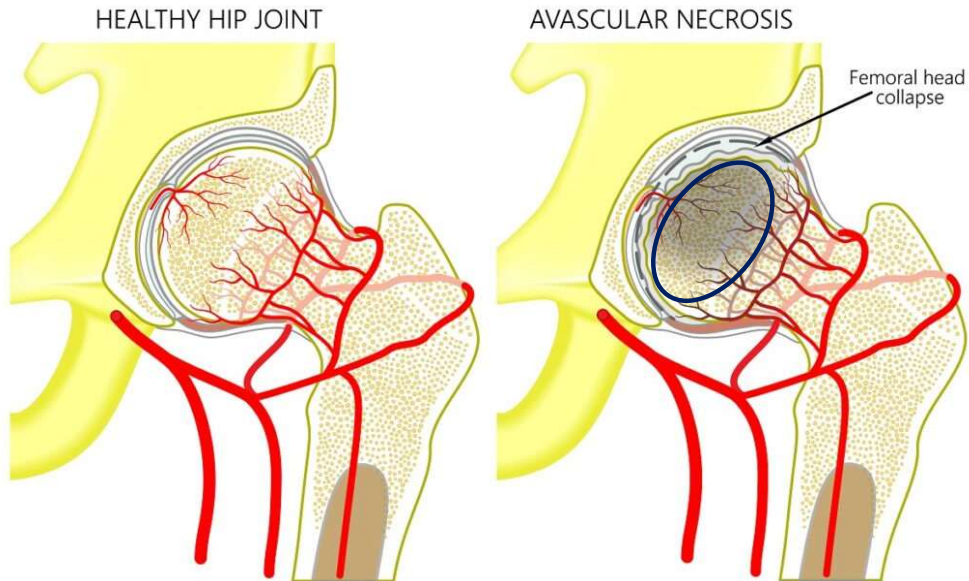
Occurs in 25% of patients who experience a femoral neck fracture

Peak age 30-50

More common in men



Pathophysiology — Reduced Blood Flow



Risk Factors

Mechanical

- Venous congestion
- Thrombotic/embolic event
- Trauma
- RBC clumping (hemoglobinopathies)

Imbalance in bone turnover

- Gaucher's disease (also increased lipid buildup)

Corticosteroids

- Inhibit fibrinolysis

Oxidative Injury

- Alcohol

Increased atherosclerosis

- Alcohol
- Corticosteroid

Hypercoagulable states

- SARS (including covid-19)
- Polycythemia

Cancer/chemo/radiation

- Hypercoagulable
- Tx causes increased bone density

Auto-immune conditions

- Increased inflammation

Metabolic disease

- Diabetes
- Liver disease



Hemoglobinopathies – ie thalassemia, sickle cell, etc.
Polycythemia



Usually age 4-10, idiopathic
Brace or cast to try to maintain round femoral head

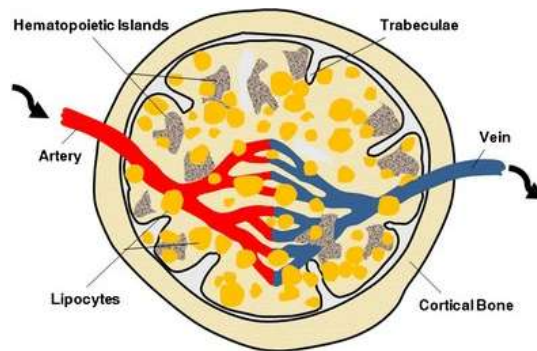
OSTEONECROSIS - Clinical

Pain

- Insidious with steroid use
- Acute with Gaucher's disease, infarctions

Limited motion

- Late symptom
- progressive



Present with pain and often need surgery
Gaucher's disease shows up on boards

Usually have limited motion as time goes on

OSTEONECROSIS DIAGNOSIS

History and Physical Examination

- Often insidious start
- Hx of tobacco, alcohol, IV drug use
- Trauma
- Point tenderness over bone

Imaging Studies

X-ray – late findings

- Crescent sign
- Sclerosis

CT

Bone scan

MRI – gold standard



Trauma, injury to bone, point tenderness over bone. Tobacco, alcohol...
Really do need imaging. MRI picks up at an earlier stage



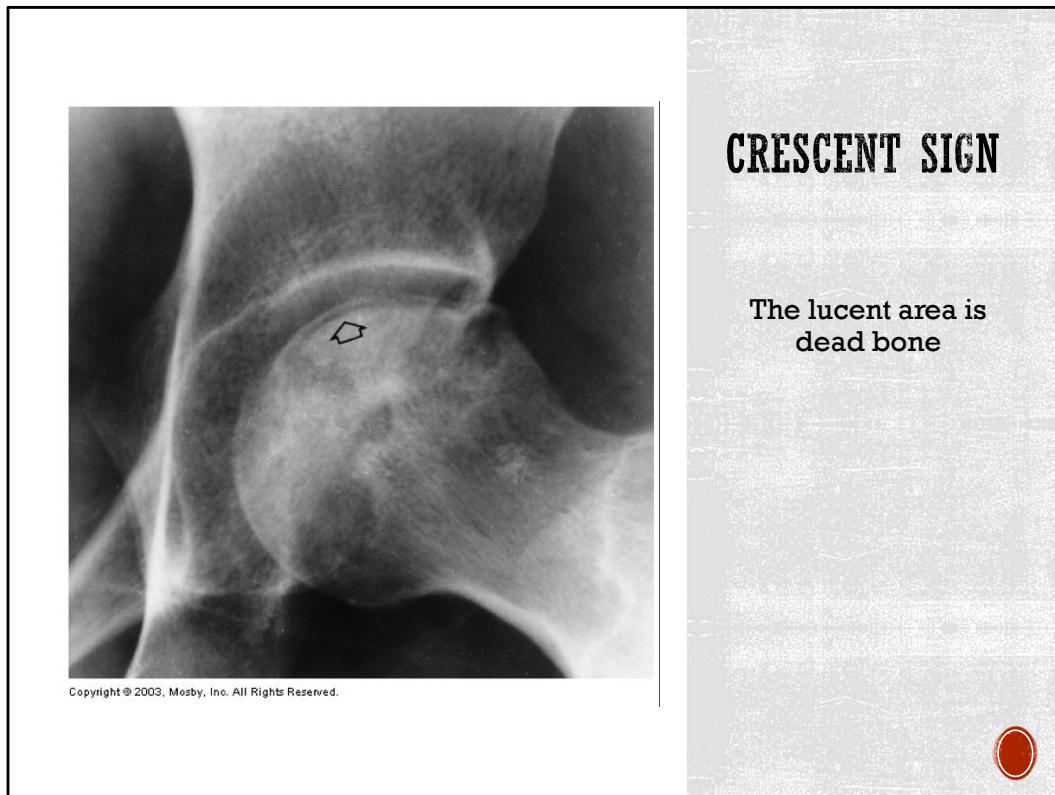
Bone Sclerosis

Living bone reacts
to necrotic area
and tries to repair
it by inducing
inflammation

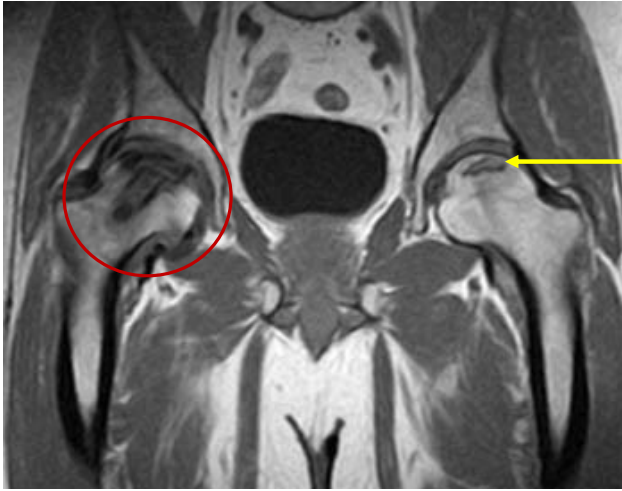


Bony sclerosis at head – more opaque – whiter

Living bone reacts to and tries to repair necrotic area by inducing inflammation



Crescent sign – crescent that is lucent – dead bone



MRI

Gold standard

Detects disease earlier

This patient has a history of AVN with compression in the R hip. The MRI diagnosed an asymptomatic AVN in the L hip



OSTEONECROSIS - Treatment

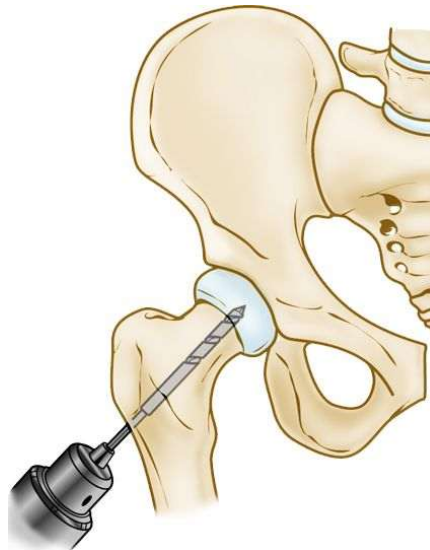
No weight bearing

- If caught early enough
- Bone is soft and easily ground down with weight

Symptom Control

Surgery

- Core decompression
- Bone and cartilage grafting
- Joint replacement usually required



Bone dead due to poor blood,
Blood softens and compresses when weight bearing so no weight bearing. Grinds away and destroys joint.

Usually by time have pain, have damage – primary tx is surgery.
Core decompression is often combined with bone and cartilage grafting to help regenerate healthy bone and support cartilage at the hip joint.
Mostly bone grafting and resurfacing as a temporary measure.
Usually need a joint replacement

Osteomalacia and Rickets

* BONE SOFTENING

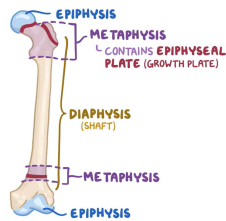
↳ FAULTY PROCESS of BONE MINERALIZATION

↳ DEFICIENT or IMPAIRED METABOLISM of **VITAMIN D**

PHOSPHATE
CALCIUM



RICKETS
(in CHILDREN)

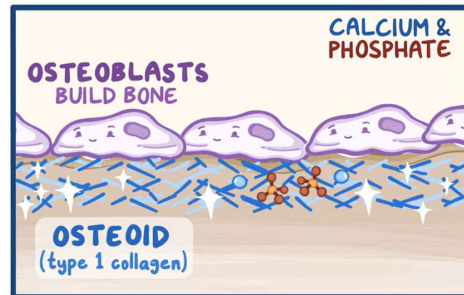


OSTEOMALACIA
(in ADULTS)

OSMOSIS.org
2022 Edition

Bone Basics

* BONE MINERALIZATION



Osteoblasts secrete osteoid to form a matrix

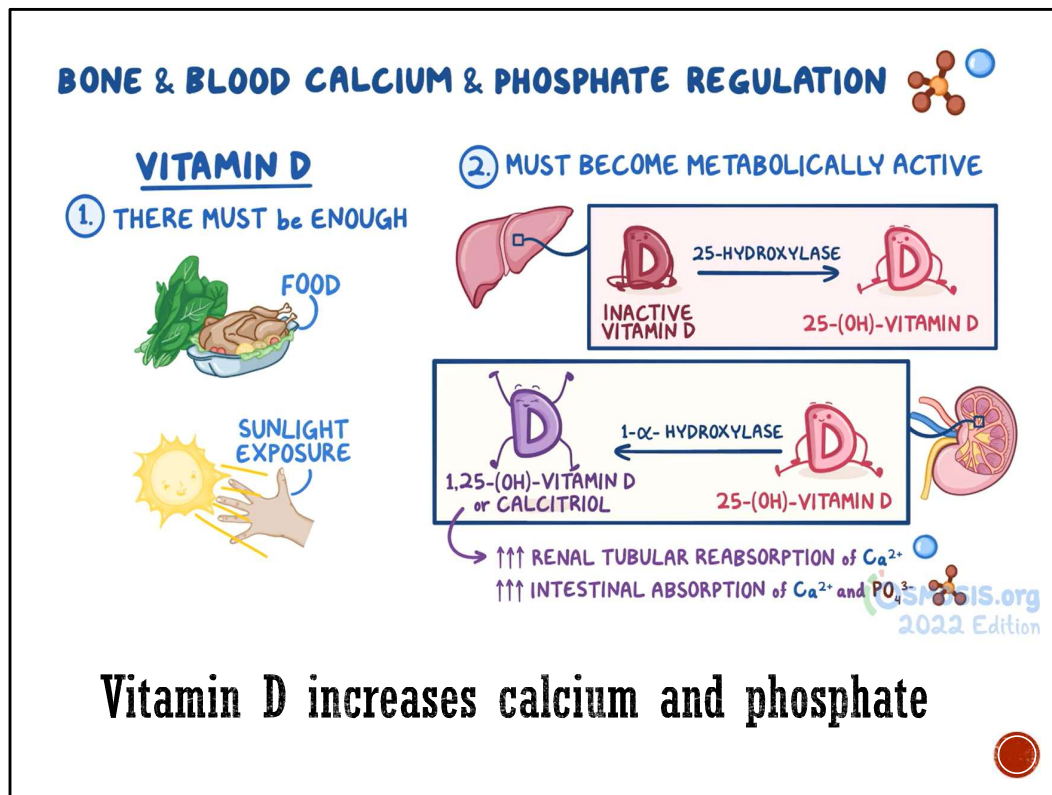
When osteoblasts start working, alkaline phosphatase is activated and stimulates uptake of calcium and phosphate

Osteoblasts are able to deposit calcium and phosphate into the bone to fill in the matrix

Deposited form is hydroxyapatite $\text{Ca}_5(\text{PO}_4)_3(\text{OH})$, not just regular calcium phosphate

OSMOSIS.org
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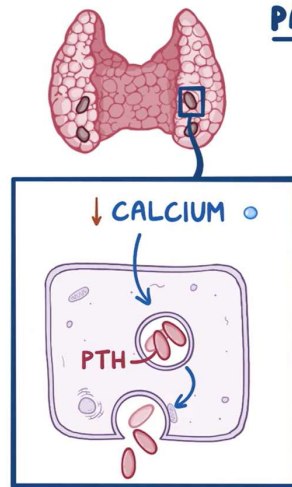




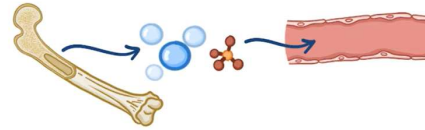
Sunlight provides cholecalciferol (not very active form of vit D3)
 Food provides Vit D2 and D3 (inactive)
 Liver converts all to calcidiol (inactive D3)
 Kidney converts to calcitriol (active)

PTH Regulates Ca, Vitamin D and Phosphate

PARATHYROID HORMONE



* STIMULATES RESORPTION of Ca^{2+} and PO_4^{3-}



* BOOSTS 1- α -HYDROXYLASE ACTIVITY

↑↑↑ ACTIVE VITAMIN D

↑↑↑ GUT ABSORPTION of Ca^{2+}

↑↑↑ Ca^{2+}

↓↓↓ PO_4^{3-}

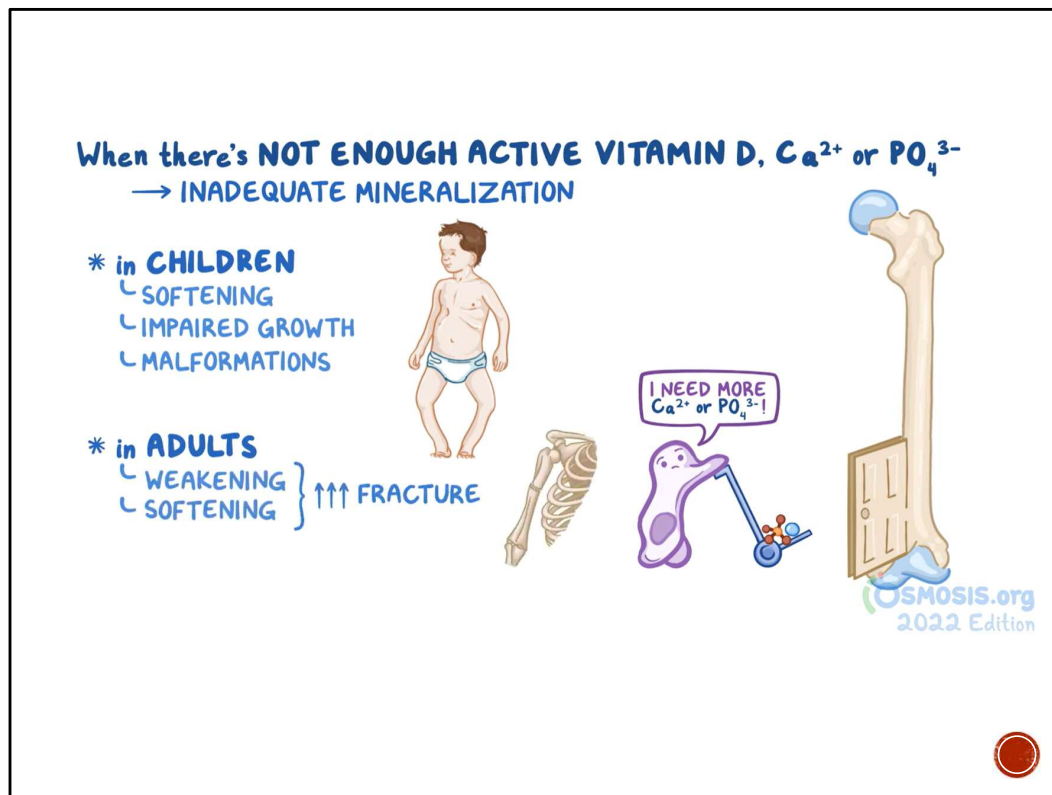
} REABSORPTION from KIDNEYS

↑↑↑ PO_4^{3-} EXCRETION

OSMOSIS.org
2022 Edition

PTH increases Ca but decreases phosphate





Both osteomalacia and rickets are due to poor mineralization, particularly with reduced vitamin D and/or Ca.

Rickets occurs in growing children and refers to inappropriate mineralization at the growth plates – they are not able to fully mineralize so they seem wide on x-ray.

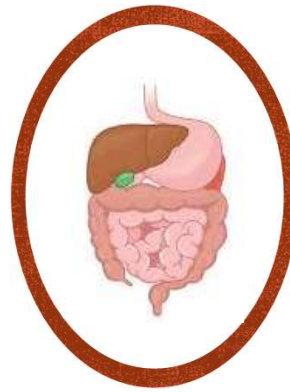
Frontal bossing occurs in children as osteoid is continually laid down but then can't be mineralized.



Diet



Drugs/
Medications



Physical/
Environmental

OSTEOMALACIA - Causes



OSTEOMALACIA - Causes



Diet

Most common cause is lack of Vitamin D

- Usually this is due to inadequate intake or production
- There are a couple of types of “vitamin D resistant rickets” caused by problems in the hydroxylation pathway that produces active vitamin D

Lack of calcium rare in children, not conclusive if causes in adults

Lack of phosphorous very rare – it is ubiquitous in foods



OSTEOMALACIA DRUGS/MEDICATION



Seizure medications can inactivate vitamin D in the body

- Dilantin
- Tegretol

Glucocorticoids can prevent the absorption of vitamin D in the gut



Glitazones can decrease stem cell differentiation to osteoblasts and decrease osteoblast activity

Vitamin D abnormalities from sz med – vitamin D turned into other forms that are not metabolically active

OSTEOMALACIA

Physical/Environmental



Poor Absorption

- Celiac disease – due to intestinal damage poor absorption
- Small intestine bypass
- Removal of the small intestine

Renal disease

Liver disease

Environmental

- Lack of sunlight

How can each of these factors predispose someone to osteomalacia?
(answers in slide notes)

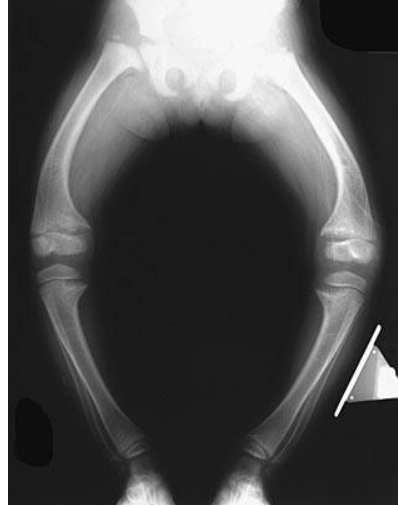


Vitamin D is absorbed in the small intestine, particularly the ileum and jejunum. Inflammation from celiac disease will prevent absorption. Bypass or removal of intestine will greatly diminish any absorption.

Remember that kidneys convert the inactive form of vitamin D to active. In renal disease, that is compromised

The liver is the first step of converting vitamin D from the sunlight or from the diet into 25-hydroxyvitamin D3 which is then converted to the active form in the kidney

Physical Findings



Physical Findings

Rachitic Rosary:
prominent nodules at
the costochondral joints

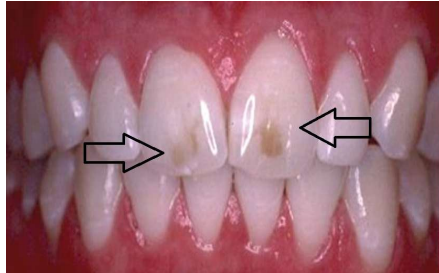


Pectus Carinatum:
protrusion of sternum
(pigeon chest)



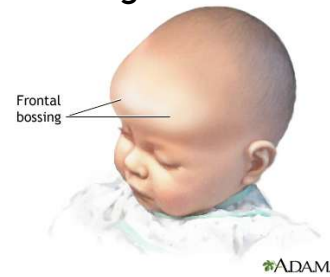
Decreased cartilage mineralization of open epiphyseal growth plates leads to enlargement of the the costochondral junctions

Physical Findings

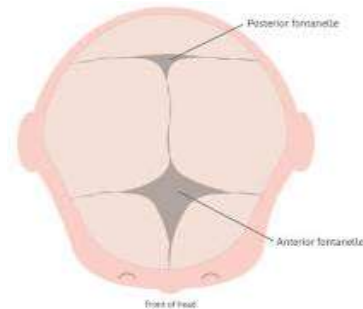


Dental Hypoplasia

Frontal Bossing



The skull is soft (craniotabes) and fontanelles close later



Putting down extra osteoid (mostly collagen 1) without mineralization so enlarges the front of the skull

OSTEOPOROSIS VERSUS OSTEOMALACIA

Feature	Osteoporosis	Osteomalacia
Osteoid	Reduced bone density (porous) with normal mineralization content (hydroxyapatite)	Often low bone density. Marked by abnormal mineralization composition making bone soft
Pathophysiology	Decrease in osteoblast to osteoclast activity	Almost always caused by very low vitamin D. Rarely, low Ph or Ca
Vertebral fractures	May be present	Usually absent
Symptoms	Silent disease	Bone pain, muscle weakness (especially thighs), waddling gait, bone deformities
Biochemical picture	Usually normal calcium, phosphate and bone-specific alkaline phosphatase. Vitamin D may be high/low/normal.	Vitamin D is usually very low. Associated with a high PTH, hypocalcemia, hypophosphatemia and a high alkaline phosphatase.
Treatment	Prevent falls, bisphosphonates	Regulate vitamin D level



Chart adapted from MyEndoConsult:

<https://myendoconsult.com/learn/osteomalacia-vs-osteoporosis/>

PAGET DISEASE OF THE BONE

A localized bone disorder which causes increased bone remodeling

Increased Osteoclastic activity

Incidence

- 1-3 million in the United States
- 2.5% of men
- 1.6% of women
- Varies some by geography

Unknown etiology – may have some genetic components

25-40% have a family history of it



4 Phases of Paget Disease

Lytic phase

- Increased osteoclasts (100x normal)
- Enlarged osteoclasts

Mixed lytic and blastic phase

- Rapid bone formation
- Increased normal appearing osteoblasts

Sclerotic phase

- Predominantly bone formation
- Woven bone – disorganized pattern/structure

Burned out phase

- minimal or absent osteoblastic and osteoclastic activity



1. Overwhelming number and outsized osteoclasts
2. Osteoblasts try to compensate by becoming more active – appear normal. Body trying to heal itself
3. Osteoblasts can't lay down matrix so fast in correct format so see a disorganized structure
4. Both blasts and clasts subside - burnout

PAGET DISEASE OF THE BONE

Clinical

Asymptomatic 75%

- Often found incidentally on x-ray or bone scan
- Elevated alkaline phosphatase seen on lab
- Usually no treatment needed in these patients

Symptomatic 25%

- Fractures
- Bone pain
- Increased warmth
- Secondary Osteoarthritis
- Neurologic (cranial nerve, vertigo, tinnitus)



Symptoms:

Lytic areas of bone and then abnormal formation

Bone pain due to rapid growth of bone by osteoblasts stretches periosteum, where there are many pain fibers

Increased warmth due to rapid turnover

If areas of growth are in joints, cause arthritis

Neurologic depending on where activity is

PAGET DISEASE

Diagnosis



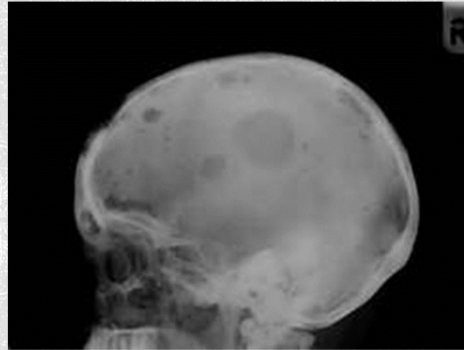
Bone Scan

Increased activity

X-rays

Lytic lesions, sclerosis,
cortical thickening, bone
enlargement

Bone Specific Alkaline
Phosphatase Elevated

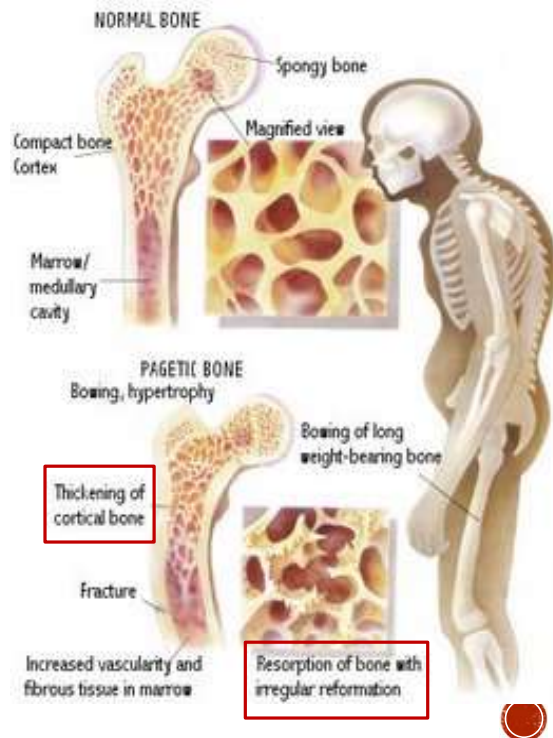


PAGET DISEASE

Diagnosis



Increased
fracture risk



Elevated bone AlkPhos due to high osteoblast activity.

Cortical thickening coarse trabecular markings from disease activity

sclerosis from extra blast activity

Lytic lesions from destruction as well as formation

PAGET DISEASE - Treatment

NONE if localized and asymptomatic

- Not going to develop long-term problems – fractures, arthritis

Indications for treatment

- Symptomatic (pain, radiculopathy, compression of optic nerve...)
- Fractures
- High disease activity with cardiac risk (elevated Ca)
- Prior to otherwise planned surgery on affected bone

Medications

- Bisphosphonate
- Calcitonin

MOA: Reduce
osteoclast activity



Bisphosphonate is first-line as it works by reducing osteoclast activity. Calcitonin can also work – does have some long term concerns such as increased risk of cancer

<https://academic.oup.com/jcem/article/102/9/3111/3892399>

Medscape emedicine

Indications for drug treatment of Paget disease are as follows ^[33] :

- Metabolic active disease - Bone pain, fatigue fracture, skull/spine fracture, radiculopathy, osteolytic lesions, bony deformities, weight-bearing bone involvement, compression of spinal cord and nerve roots, bone compression of the eighth cranial or optic nerve
- Preparation for orthopedic surgery (joint replacement anticipated at involved sites within 6 months)
- Hypercalcemia or hypercalciuria - Recurrent renal calculi due to [hypercalciuria](#), or fractures; serum alkaline phosphatase or urine hydroxyproline levels greater than twice the upper limit of the reference range; immobilization

PAGET DISEASE - Prognosis

Asymptomatic/localized

EXCELLENT

Diffuse with treatment

GOOD

Patients who develop
sarcoma due to the
increased bone
remodeling have a

5% FIVE-YEAR SURVIVAL

Juvenile Paget Disease

POOR

SARCOMA Types

Angiosarcoma

Malignant neoplasm
in the vessel walls

Osteosarcoma

Tumor in a bone

Ewing's sarcoma

Bone

Chondrosarcoma

Cartilage

Gastrointestinal stromal tumor

Mesenchymal neoplasms
of the gastrointestinal tract

Liposarcoma

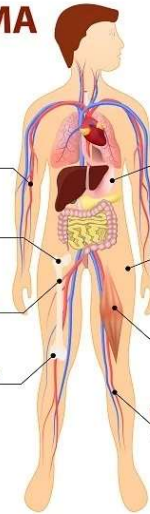
Fat cells

Fibrosarcoma

Fibrous connective tissue

Hemangioendothelioma

Vascular neoplasms



PAGET DISEASE - Juvenile Form

- Rare disease caused by TNFRSF11B gene mutation
- Autosomal recessive
- Begins in Infancy
- Presents with generalized bone disease
- Bones enlarge and deform
- Bones easily fracture
- Decreased life expectancy - rare patients survive to 50



PAGET DISEASE - Juvenile Form

Bone pain

Dwarfism

Failure to thrive

Pigeon chest

Muscular weakness

Loss of teeth

Progressive enlargement of
the head

Bowing of the long bones of
the legs



Longest surviving patient with juvenile paget disease – 60 year old man from Greece. Has been on bisphosphonate tx for 2 decades. Has suffered from progressive deafness and is legally blind. Research biologist who was diagnosed formally during his PhD studies.

<https://asbmr.onlinelibrary.wiley.com/doi/full/10.1359/jbmr.070307>

OSTEOPETROSIS

Opposite of Paget Disease:
caused by the failure of
osteoclasts to resorb bone

“Marble bone disease”

Bone is dense but still prone
to fracture

Overgrowth of cortical bone
fills marrow space

- Pancytopenia
- Can impinge on nerves and
result in palsies

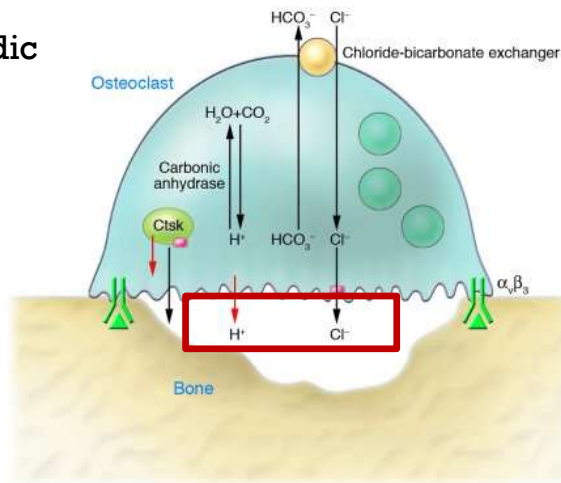


OSTEOPETROSIS

Osteoclasts need an acidic environment to resorb bone.

Multiple etiologies that may not provide the correct environment for osteoclasts to work

Common mutation:
Carbonic anhydrase
isoenzyme II deficiency



Acidic environment allows osteoclast to attach to bone as well as to break down the hydroxyapatite

<https://www.jci.org/articles/view/28605/figure/3>

Osteopetrosis - Presentation

Bones are fragile and can fracture easily

Nasal stuffiness due to mastoid and paranasal sinus malformation is often the presenting feature of infantile osteopetrosis

Neuropathies related to cranial nerve entrapment occur due to failure of the foramina in the skull to widen completely

Extra osseous tissue replaces bone marrow

- Pancytopenia – anemia, bleeding and recurrent infections

Other head manifestations include deafness, optic atrophy, and hydrocephalus



<https://emedicine.medscape.com/article/123968-clinical>

OSTEOPETROSIS - Treatment

The earlier the diagnosis, the poorer the prognosis

No effective treatment -Watch for and treat symptoms

Three forms:

Infantile (recessive)

Erythropoetin
Bone marrow transplant

Pediatric (recessive)

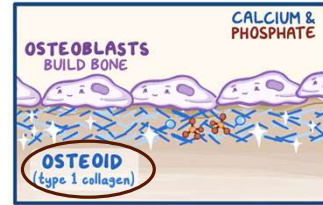
Surgery for fractures

Adult (dominant)

Generally no treatment



OSTEOGENESIS IMPERFECTA



Genetically based connective tissue disease

Called “brittle bone disease”

Caused by inability to produce enough type 1 collagen

Etiology – most common type is autosomal dominant

COL1A1 gene (chromosome 17)

COL1A2 gene (chromosome 7)

Relatively rare with incidence 1 per 20,000

But, highly tested!



OSTEOGENESIS IMPERFECTA - Clinical

Fracture easily

- Can fracture during the birth process
- Before diagnosis, children can come in with multiple fractures that don't heal easily and may seem like child abuse

Blue sclerae present due to the thin connective tissue showing the choroidal veins under them

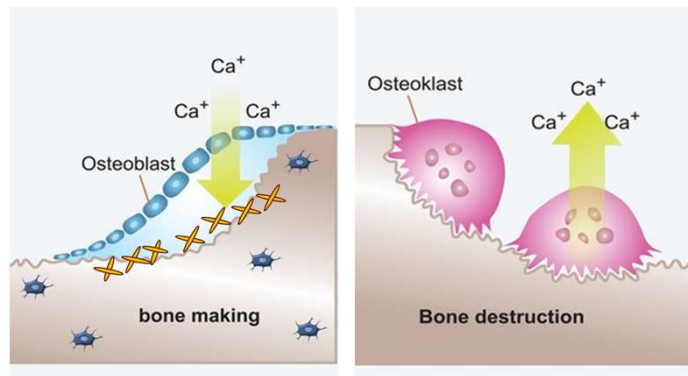
Hearing loss due to abnormal ossicles

Dentin of teeth wears down easily leaving imperfections

Shortened stature



Recap...



1. Reduced osteoblast to osteoclast activity _____
2. Reduced osteoclast activity _____
3. Increased osteoclast activity _____
4. Lack of vitamin D _____
5. Poor blood supply to bone _____
6. Lack of type collagen I production _____

